

AI Study Assistant Agent:

Problems in the Agent Context:

The main Problem is to help students study more effectively. The agent needs to understand the Students Subjects, Study Progress, quick result, week topics and available study time. Based on this information, it can create a personalized Study Plans, recommended topics, generate Practice questions & track the Students Progress.

Problem Formulation:

Textual Formulation:

States:

The Student's current Progress, completed topics, quiz scores, weak areas and available study time.

Initial State:

The student enter their subject, topics, previous Performance and available study time.

Actions:

Create a Study Schedule, recommended topic generate Practice questions, give quizzes, Provide explanation and update the Student's Progress.

Transition Model:

When the Student Complete a topic or quiz, the agent update the students Progress and adjusts the Study Plan according to their Performance.

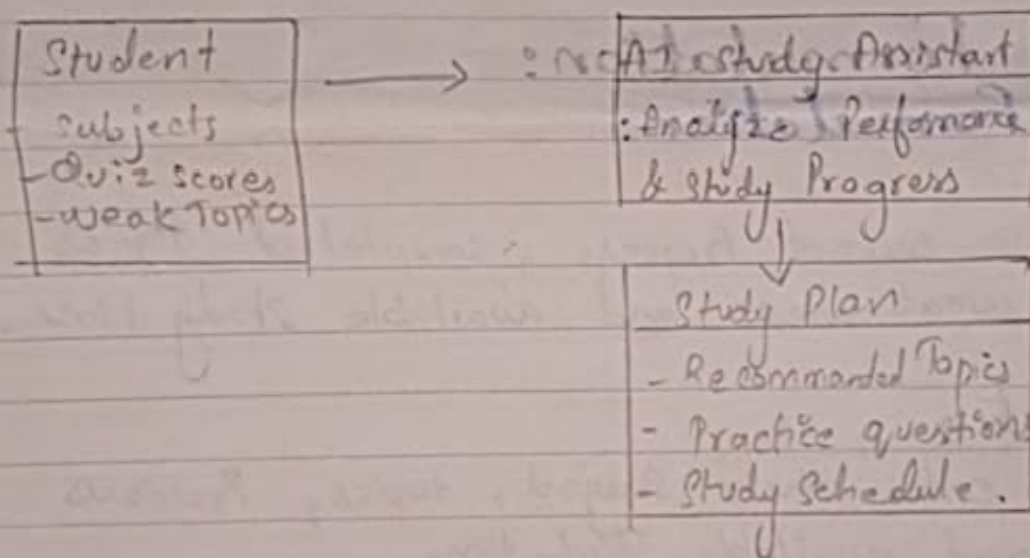
Goal Test:

The student completes the required topics and improve their performance before the target exam or deadline.

Path cost:

Extra study time, repeated topics and unnecessary practice increase the cost. The agent aims to achieve better learning outcomes using the student's available time efficiently.

Graphical Formulation:



PEAS Evaluation

1) Performance Measure:

Improvement in quiz scores, completion of study goals, effective use of study time, and student progress.

2) Environment :-

Student, subjects, study materials, quizzes, assignment, exams and learning schedules.

Actuators:

Study Plan Recommendations, notification, explanation, quizzes and topic recommendations.

Sensors:

Quiz-scores, student responses, completed topics, studytime and feedback from the student

Environment Properties:

Partially observable

Stochastic

Sequential

Dynamic

Discrete

Single Agent